

ORIGINAL ARTICLE

Can computerized working memory training improve impaired working memory, cognition and psychological health?

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Abstract

Objective: To study if computerized working memory (WM) training, in the sub-acute phase after acquired brain injury, in patients with impaired WM, improves WM, cognition and psychological health.

Research design: A randomized study ($n=47$) with an intervention group (IG) and a control group (CG), mean age 47.7 years. The WAIS-III NI, Digit span, Arithmetic, Letter-Number Sequences (Working Memory sub-scale), Spatial span, the Barrow Neurological Institute Screen for Higher Cerebral Functions (BNIS) and the self-rating scales DEX and HADS were administered at baseline and at follow-ups at 6 and 18 weeks. Both groups underwent integrated rehabilitation. The IG also trained with the computerized WM training program, Cogmed QM, which was offered to the CG and followed up after the study completion.

Results: Both groups improved after their WM training in Working Memory, BNIS and in Digit span, particularly the reversed section. Both the BNIS and the Digit span differed significantly between the IG and CG due to the greater improvement in the IG after their WM training. Psychological health improved as both groups reported less depressive symptoms and the CG also less anxiety, after the training.

Conclusion: Results indicated that computerized WM training can improve working memory, cognition and psychological health.

Keywords

Acquired brain injury, cognition, computerized training, psychological health, rehabilitation, working memory

History

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Introduction

Working memory (WM), which is often impaired after brain injury [1, 2], includes the temporary storage and processing of information that is necessary for cognitive activity [3]. Miller [4] described already in 1956 the short-term memory (STM) by the magical number 7 ± 2 as some limits on the capacity for processing information. In 1974, Baddeley and Hitch [5] proposed to divide the unitary STM into three sub-systems assumed to work together as part of a unified WM system for performance of complex tasks; an attention control system and the central executive, supported by subsidiary storage systems for phonological (phonological loop) and visuospatial (visuospatial sketchpad) information [6, 7]. WM also filters irrelevant information and is a prerequisite for sequential information processing in logical reasoning and problem-solving, which uses previously acquired information to compare different outcomes while still maintaining focus. As an executive function with a central role in cognition, WM may be observed as brain activity specifically in the frontal and parietal cortex [8–12]. Deficits in WM affect a person in

most activities, as problems manifest in impaired attention, planning [1], maintenance and organization of information, problem-solving and execution of actions. Impaired WM results in memory disturbances as to what to do next, in which one ‘loses the plot’ and has difficulties in finding the context of a situation [1, 6, 7]. Consequently this causes major issues in the capacity to perform duties that had been easy to perform prior to injury [13].

Recent studies point towards the potential to improve WM and that WM training can be generalized to affect non-trained tasks [14]. Increased prefrontal and parietal cortical activity and changes in the biochemistry of the brain after training have also been reported [15].

The computerized WM training program, RoboMemo, was developed in Sweden for children and adolescents diagnosed with ADHD [16]. The outcomes of this program suggest that WM can be trained and that an improvement could increase problem-solving ability, school performance and quality-of-life [16].

The computerized training program Cogmed QM, earlier known as ReMemo, was designed for adults with ADHD [17]. When used for patients with acquired brain injury, general attention was shown to improve at 1-year post-injury [13, 18]. In addition, short- and long-term improvements in neuropsychological WM-test results were reported, even when training

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occurred 1–3 years after the injury [19]. The results of these studies suggest that training effects also led to general improvements in cognitive and executive functioning in daily life, increased activity, participation and a better quality-of-life [20].

Working memory, fundamental for higher cortical functions and functioning in daily life, is often dysfunctional after brain injury. It is, therefore, of great importance to explore if computerized WM training enhances WM and improves the efficiency of early rehabilitation. This might also be supported by Cramer [21], who reported that early stimulation after a brain injury may increase the potential for the inherent plasticity of the brain to repair the damage.

Aims

The aim of this study was to investigate whether patients with a dysfunctional working memory improve working memory, cognitive function and psychological health by computerized training with the Cogmed QM program.

Method

Study design

The study was a randomized, controlled study including all outpatients at the Department of Rehabilitation Medicine, Sahlgrenska University Hospital, Göteborg, Sweden, during March 2008 to December 2010.

In the study sample it was decided to include patients below normal range (6 ± 1) with a performance of ≤ 5 digits forwards and/or ≤ 4 digits reversed in the WAIS-III Digit span and/or ≤ 5 blocks forwards and/or ≤ 4 blocks reversed in the WAIS-III NI Span board [22, 23]. Exclusion criteria were aphasia/non-Swedish communicable or if participation in the study was contraindicated due to medical reasons according to the physician's health assessment (i.e. pronounced fatigue, pain or depression). All approved patients received verbal and written information about the study and their right to refrain from further participation. Patients that agreed to participate submitted written informed consent.

After a baseline assessment (A1) the patients were randomized by lot into either the intervention group (IG) or the control group (CG) (Figure 1). Both groups received rehabilitation in accordance with the normal routines at the clinic, based on their rehabilitation needs. The IG also trained for 5 weeks with the computerized WM training program, Cogmed QM, during their rehabilitation period.

Both IG and CG were assessed after the 5-week training program for the IG (A2) and also followed-up 3 months later (A3). Due to ethical reasons, the CG was then offered the same Cogmed QM training and followed-up at a last final assessment (A4).

As only eight of the 18 patients in the CG returned for a WM training period, the group was split into the C1 ($n = 8$), who accepted the WM training, and the C2 ($n = 10$), who refused the training. Differences between these sub-groups were examined at all assessments.

The study was performed within the ordinary clinical work as a development effort of a treatment method at the Rehabilitation department with permission from the head of the department and has, therefore, not been considered by an ethical committee. Ethical consideration was made throughout to secure participant safety and integrity.

Instruments

Demographic data were collected from the patients' medical charts.

The level of consciousness was assessed using the Reaction Level Scale (RLS85) [24]. The difference from Glasgow Coma Scale (GCS) is that the RLS85 only takes into account the motor response at the examination of the patient with a reduced wakefulness. Starmark et al. [25] have shown the rank correlation between the GCS sum score and the RLS85 to be -0.94 , suggesting the same ranking order of severity and indicating that the underlying concepts of somnolence, delirium and motor responses in coma are evaluated in the same way. The mentally responsive part of RLS includes the scores 1–3 and the scores 4–8 assess mentally not responsive and unconscious, where a score of 8 also includes no response to pain stimulation (Table I).

Table I. Reaction level scale (RLS85) clinical descriptor responsiveness score.

Alert	No delay in response	1
Drowsy or confused	Responsive to light stimulation	2
Very drowsy or confused	Responsive to strong stimulation	3
Unconscious	Localizes but does not ward off pain	4
Unconscious	Withdrawing movements on pain stimulation	5
Unconscious	Stereotype flexion movements on pain stimulation	6
Unconscious	Stereotype extension movements on pain stimulation	7
Unconscious	No response on pain stimulation	8

Starmark et al. [25].

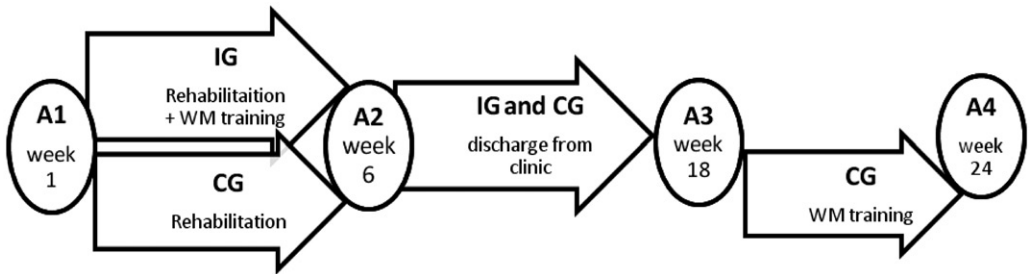


Figure 1. Study design. Assessments 1–4 (A1, A2, A3 and A4) for the Intervention Group (IG) and the Control Group (CG).

Digit span and Span board, sub-tests of the Wechsler Adult Intelligence Scale III (WAIS-III NI), are comprised of two parts each, immediately repeating a verbal and spatial series, both forwards and reversed. Lezak [22] reports the normal range for digits forwards as 6 ± 1 as normal. Also 'spans of 6 or better are well within normal limits, a span of 5 may be marginal to normal limits, a span of 4 is definitely borderline and 3 is defected' (p.359). Corresponding performance on the Span board series is 6 ± 1 [26]. In addition to the Digit span and Span board the comprised sub-scale of Working Memory, aggregated from the combined scaled-scores of the three tests Digit span, Arithmetic and Letter-Number Sequences of the WAIS-III NI [27] was used. A maximum score is 57 points. These tools were used to measure working memory and information processing [23].

The Barrow Neurological Institute Screen for Higher Cerebral Functions (BNIS) is a cognitive screening instrument consisting of seven sub-scales: speech and language, orientation, attention/concentration, visuospatial and visual problem-solving, memory, affect and self-awareness vs performance [28–30]. A pre-screen outcome determines if the patient is testable. The result is reported as a composite score with a maximum of 50 points. A cut-off score <47 points indicates a cognitive dysfunction.

The Dysexecutive Questionnaire (DEX) is a self-report questionnaire measuring experienced dysexecutive problems in emotion, personality, motivation, behaviour and cognition to assess problems in goal-directed behaviour [31]. Twenty items, five for each factor (inhibition, intentionality, knowing/doing dissociation, in-resistance social regulation), are scored on a 5-point scale [32]. Higher scores yield more dysexecutive problems; a score ≥ 27 points is considered as a cut-off for severe or very severe difficulties in executive function [37].

The Hospital Anxiety and Depression Scale (HADS) is a self-assessment questionnaire that measures emotional and psychological health [33]. Each question is scored between 0–3 points. For both anxiety and depression a score of 8–10 indicates a mild, 11–14 points a moderate and scores >15 points a severe condition [34].

The computerized WM training program Cogmed QM [35] was used according to the guidelines, i.e. 30–45 minutes, 5 days/week for 5 weeks. The program includes a battery of visuospatial and verbal auditory WM tasks and all parts of the battery must be trained at each session, 90 trials each day. All tasks involve; (i) maintenance of multiple stimuli at the same time, (ii) short delays during which the representation of stimuli should be held in WM, (iii) unique sequencing of stimuli order in each trial and (iv) the difficulty level adapting as individual performance [13]. Lundqvist et al. [19] described the different tasks of the program; the visuospatial WM tasks require remembrance of the position of stimuli in a four-by-four grid and then reproduction of the stimuli in the same order, in the reverse order or in a rotated grid. The verbal WM tasks require remembrance of sequences of letters and digits forwards and/or backwards. Each task is introduced by a speaker voice and the person responds by localizing and remembering the stimuli. The software directly includes reinforcement which was implemented via scores and positive verbal feed-back [13, 18]. Personal and individual feedback once a week, by a specially trained coach, was also included.

Statistical methods

Descriptive statistics are presented as frequencies and/or percentages, as means (*M*) and standard deviations (SD) and/or medians (Md) and ranges. Non-parametric statistics were used to test differences between the groups (Mann-Whitney U-test, independent samples, for ordinal data and Chi-square test for nominal data) and differences within groups between different assessment occasions (Wilcoxon signed-rank test). Spearman's Rank correlation, rho, was used to examine associations between variables. Significance levels of ≤ 0.05 were accepted. The statistical package SPSS (version 15.0) was used.

The analyses consisted of comparisons between the IG and CG at each assessment, for relevant variables, and also within each group between different assessments. Comparisons between the assessments were based on comparing the calculated changes in scores between the assessments within each group.

Results

Background

During the study 331 patients were admitted to the outpatient rehabilitation at the clinic. Of these, 47 patients out of 121 identified with an impaired WM became part of the study (Figure 2).

Thirty-eight patients completed the three assessments according to the study design, 20 in the IG and 18 in the CG (Table II).

After the third assessment (A3) (Figure 1), when all the patients were discharged from the clinic, the CG was invited for Cogmed QM training, of which 11 patients accepted but only eight completed. The results of these patients (C1) were compared with the remaining 10 patients (C2) who refused the Cogmed QM training, using the an in Mann Whitney U-test. A significant difference was found at baseline between these groups only in the depression section of HADS ($p = 0.020$).

Baseline data (A1)

There were 24 men (51%) and 23 women (49%) accepted into the study, with a mean age of 47.7 ± 11.266 (Md = 51, range = 22–63 years) (Table III).

Median time since injury was 32 weeks (12–500 weeks). Overall median RLS was 1 (1–6); in the IG Md = 1 (1–6) and in the CG Md = 1 (1–5) on arrival to the hospital. The majority, 63.8%, were fully conscious (RLS = 1) and only 8.5% were unconscious (RLS = 4–6). No significant differences were found between the IG and CG regarding gender, age, time since injury, RLS, educational level, diagnosis (Table III) or scaled scores for working memory at A1 (Table IV). Nor did RLS significantly differ between gender or diagnosis or between the control sub-groups C1 and C2.

Digit span

In the Digit span both the IG and CG improved significantly after their respective computerized WM training periods, that is the IG in A2 ($p = 0.003^{**}$) and the C1 in A4 ($p = 0.024^{*}$), as compared to A1 (Table V). Also, the whole

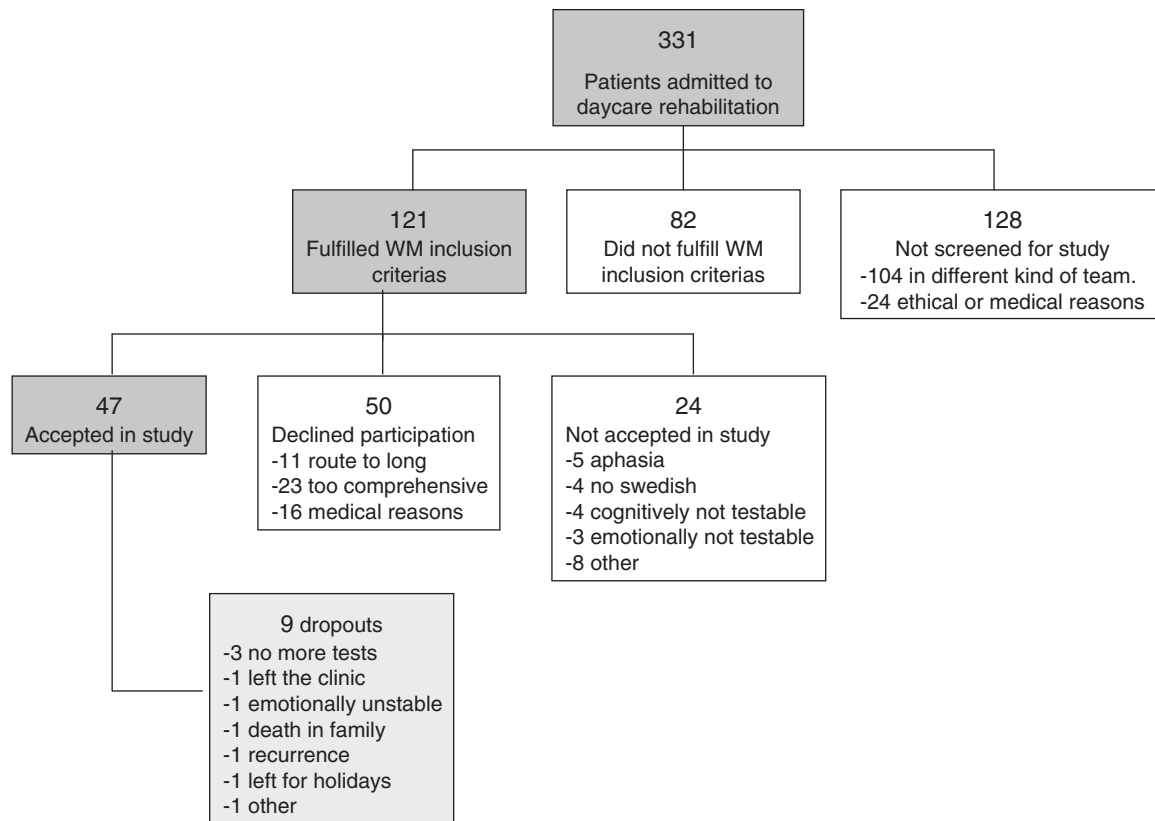


Figure 2. The study population. Flow-chart.

Table II. The number of tested participants (*n*) in the Intervention Group (IG) and the Control Group (CG) at the four neuropsychological assessments, A1–A4. The difference in numbers is due to dropouts during the study.

Assessments	IG (<i>n</i>)	CG (<i>n</i>)
A1 – baseline	25	20
A2 – after training (IG), week 6	22	18
A3 – week 18	20	18
A4 – after training (C1), week 24	–	8

CG improved in Digit span at A3 ($p=0.014^*$). Further analysis showed that the improved ability in the reversed section affected the total result of the Digit span (Table IV). After their respective WM training, the IG improved significantly in the reversed section between A1 and A2 ($p=0.004^{**}$) and stabilized in A3 ($p=0.001^{**}$) and the C1 improved between A3 and A4 ($p=0.039^*$) (Table V). There was also a significant difference between the IG and the CG in the change in Digit span from A1 to A3 ($p=0.045^*$), after the WM training for the IG, where the IG showed a larger improvement, especially in the reversed part ($p=0.003^{**}$) (Table V).

Span board

In the Span board no significant changes were found for the IG throughout the study. The mean of the scaled score was within the average of the normative value already at the A1 stage (Table IV). For the CG a significant improvement was seen between A1 and A2 even without WM training ($p=0.003^{**}$), while the IG showed a trend towards

improvement ($p=0.064$) (Table V). No significant differences were found between the two groups in either assessment (Table IV).

Working memory sub-scale

In the Working Memory sub-scale, significant improvements were seen for both the IG ($p=0.000^{**}$) and the whole CG ($p=0.003^{**}$) between A1 and A2, when both groups had ordinary rehabilitation at the clinic, with the IG also receiving training with Cogmed QM. When analysing the split CG, only the C1 significantly improved between these assessments ($p=0.011^*$) (Table V). This group also significantly improved their score after the Cogmed QM training, (A1–A4) ($p=0.011^*$) (Table V).

Cognitive function—BNIS

A significant cognitive improvement was shown in the IG after the computerized WM training from A1 to A2 ($p=0.006^{**}$), which also remained between A1 and A3 ($p=0.006^{**}$). In the control sub-group, C1, a trend towards significance in cognitive improvement was also found after their CogmedQM training (A1–A4; $p=0.053$). The change in the BNIS score significantly differed between the IG and the whole CG from A1 to A2 ($p=0.044^*$), indicating more enhancement for the IG (Table V). However, both groups remained, on average, below 47 points, indicating a cognitive dysfunction, even after their respective WM training periods (Table IV). The variance within the groups, however, revealed several patients with outcomes within the ‘normal’ basic cognitive function range.

Table III. Background data for the Intervention (IG) and Control (CG) groups.

	Gender, male/female	Age (years), Md/range	Time since injury (weeks), Md/range	RLS, Md/range	Educational level, n, years of studies	Diagnosis frequency, % (n)
IG, <i>n</i> = 25	13/12	51 22–63	32 12–135	1 1–6	3, 1–9 years 8, 9–12 years 14, 12+ years	68% stroke (17) 16% trauma (4) 16% other (4)
CG, <i>n</i> = 20	10/10	53 22–62	22.5 16–500	1 1–5	2, 1–9 years 11, 9–12 years 7, 12+ years	75% stroke (15) 15% trauma (3) 10% other (2)

The total BNIS score and the WM sub-scale correlated significantly; As determined by the participants at A1 ($\rho = 0.615$, $p = 0.000^{**}$); A2 ($\rho = 0.452$, $p = 0.003^{**}$); and A3 ($\rho = 0.703$, $p = 0.000^{**}$) as well as separately for the IG; at A1 ($\rho = 0.687$, $p = 0.001^{**}$); A2 ($\rho = 0.450$, $p = 0.035^{*}$); A3 ($\rho = 0.755$, $p = 0.000^{**}$). Corresponding correlations with the CG separately were; at A1 ($\rho = 0.292$, $p = 0.226$); A2 ($\rho = 0.434$, $p = 0.072$); A3 ($\rho = 0.593$, $p = 0.009^{**}$), showing significance only at A3. Thus, patients with better WM also showed better general cognition as determined by the BNIS.

The proportion of patients that were aware according to the sub-scale 'awareness vs performance' differed in each of the groups at A1; in the IG seven patients of 18 (33.3%) were scored as aware and in the CG nine out of 18 (50%). The split of CG into the sub-groups revealed that proportionally more patients in C2 had a good awareness; in the C2 six out of 10 patients (60%) were aware and in the C1 three out of eight (37.5%), indicating the patients of the C2 to be more realistic about their performance level.

Executive function—DEX

The IG and the CG did not differ significantly regarding self-rated dysexecutive problems (DEX) at any assessment. Additionally, there were no significant differences found in the change of perceived executive difficulties in either group throughout the study. Both groups remained close to the 27 point cut-off, indicating severe or very severe experience of executive problems (Table IV).

Severe difficulties in executive function (Md = 30.00, range = 7–52) were experienced for the C1, even after their WM training, although their average BNIS score reached a median score of 46.50 (39–50) points, where half the group received scores above the cut-off level of 47 points.

Throughout the study the DEX significantly correlated with HADS A for the whole sample: at A1 ($\rho = 0.616$, $p = 0.000^{**}$), at A2 ($\rho = 0.766$, $p = 0.000^{**}$) and at A3 ($\rho = 0.643$, $p = 0.000^{**}$). Corresponding results were likewise found between the DEX and HADS D scores: at A1 ($\rho = 0.644$, $p = 0.000^{**}$), at A2 ($\rho = 0.648$, $p = 0.000^{**}$) and at A3 ($\rho = 0.585$, $p = 0.000^{**}$). Thus, it appears that a more anxious and/or depressive mood correlated with a more dysexecutive experience. However, this association was no longer present for the IG at A2, after the Cogmed QM training.

Psychological health—HADS A

Both the IG and CG had median values below the cut-off of eight points, indicating no anxiety throughout the study diagnosed by HADS A (Table VI).

No significant differences were obtained in anxiety between the three assessments for the IG, while the C1 significantly improved ($p = 0.027^{*}$) after their WM training at A4 (Md = 7.00, range = 0–10) compared with A1 (Md = 10.50, range = 0–15) (Tables IV and V). The IG and CG did not differ significantly in anxiety at any assessment.

Psychological health—HADS D

Both the IG and CG had median results below the cut-off score for depression throughout the study. No significant change was noted in either group between A1 and A2. At A3, compared to A1, the IG showed a significantly lower score ($p = 0.005^{**}$), indicating less depressive symptoms, while no significant difference was obtained for the CG (Table V). The sub-groups C1 and C2 differed significantly at A1 ($p = 0.020^{*}$), with the patients in the C1 reporting more depressive symptoms (Md = 9, range = 1–11) compared to those in the C2 (Md = 3, range = 1–9). Compared to A1, a significant improvement was seen in the C1, after their Cogmed QM training (A4; $p = 0.024^{*}$), indicating a less depressive mood (Md = 6.00, range = 0–11). A significant difference between the IG and C1 at A2 ($p = 0.032^{*}$) indicated the IG to report less depressive symptoms after training in the Cogmed QM program.

Discussion

The aim of this study was to investigate if computerized WM training with the Cogmed QM program could improve working memory, cognitive function and psychological health in patients with dysfunctional working memory in the sub-acute phase after acquired brain injury.

The main findings of the study showed that both groups improved after their WM training in the Working Memory sub-scale, cognitive function (BNIS) and in Digit span, particularly the reversed section. The results of the IG on the BNIS and the Digit span after their WM training differed significantly compared to the results of the CG at the same time, due to a greater improvement in the IG. Also psychological health improved after their WM training as both groups reported less depressive symptoms. The CG also experienced less anxiety, after the computerized training.

The differences found between the changes in the IG (A1–A2) and in the CG (C1; A1–A4), after their computerized WM training, in the Digit span (especially in the reversed section) and the Working Memory sub-scale (Table V), indicate that the Cogmed QM training had power to improve working memory and information processing. This corresponds to previous reported results of Cogmed QM

Table IV. Mean values (*M*), standard deviations (*SD*), median values (*Md*) and range (min–max) of the neuropsychological tests and the executive self-report DEX, between the Intervention Group (IG) and the Control Group (CG) at assessments 1–4 (A1, A2, A3 and A4). Statistical analysis of the difference between the IG and CG in each assessment (*p*-value).

	A1 – baseline				A2 – 6 weeks				A3 – 18 weeks				A4 – 24 weeks			
	IG (<i>n</i> = 25)		CG (<i>n</i> = 20)		IG (<i>n</i> = 22)		CG (<i>n</i> = 18)		IG (<i>n</i> = 20)		CG (<i>n</i> = 18)		IG (<i>n</i> = 20)		CG (<i>n</i> = 8)	
	<i>M</i> (<i>SD</i>)	<i>Md</i> (min–max)	<i>M</i> (<i>SD</i>)	<i>Md</i> (min–max)	<i>M</i> (<i>SD</i>)	<i>Md</i> (min–max)	<i>M</i> (<i>SD</i>)	<i>Md</i> (min–max)	<i>M</i> (<i>SD</i>)	<i>Md</i> (min–max)	<i>M</i> (<i>SD</i>)	<i>Md</i> (min–max)	<i>M</i> (<i>SD</i>)	<i>Md</i> (min–max)	<i>M</i> (<i>SD</i>)	<i>Md</i> (min–max)
Digit span forwards amount	5.58 (1.248)	5.50 (4–8)	5.30 (1.031)	5 (3–7)	5.86 (1.153)	6† (4–9)	5.61 (1.577)	5± (3–9)	5.50 (1.150)	5↓ (3–8)	5.44 (1.097)	5.50↓ (3–7)	5.63 (1.408)	5.50† (4–8)		
Digit span reversed amount	3.64 (1.036)	4 (2–5)	4.20 (1.005)	4 (2–7)	4.71 (1.617)	5† (0–7)	4.39 (1.195)	4± (2–6)	5 (1.328)	5± (3–7)	4.39 (1.145)	4± (3–7)	5 (1.309)	5± (3–7)		
Digit span scaled score	7.12 (1.716)	7 (4–11)	7.70 (2.080)	7.50 (4–12)	8.67 (2.834)	8† (3–15)	8.44 (2.833)	8.50† (3–13)	8.33 (2.635)	8.50† (4–14)	7.89 (2.139)	7↓ (4–13)	9.50 (3.207)	9† (5–14)		
Span board forwards amount	4.96 (1.083)	5 (2–6)	5.25 (1.209)	5 (3–8)	5.71 (1.146)	6† (4–8)	5.28 (0.895)	5.50† (3–6)	5.33 (0.970)	6± (4–7)	5 (1.138)	5↓ (3–7)	6.13 (1.126)	6† (5–8)		
Span board reversed amount	4.71 (1.268)	4 (2–7)	4.65 (1.182)	4.50 (3–9)	5.40 (1.392)	6† (4–8)	4.56 (0.984)	4↓ (3–6)	5 (1.188)	5↓ (3–7)	4.61 (1.243)	5† (2–6)	5.13 (0.835)	5± (4–6)		
Span board scaled score	8.96 (3.507)	10 (2–14)	8.20 (2.505)	8 (3–13)	10.45 (3.591)	11† (4–16)	9.33 (2.765)	10† (3–14)	9.44 (3.294)	9.50↓ (4–16)	8.83 (2.618)	10± (3–12)	11.13 (2.696)	11† (8–16)		
WAIS-III NI working memory scaled score	22.52 (7.012)	24 (9–33)	25.84 (6.203)	28 (10–33)	27.52 (6.846)	28† (15–41)	28.44 (6.879)	30† (12–38)	26.61 (8.103)	25↓ (14–44)	26.78 (6.975)	29↓ (13–36)	30.38 (7.726)	34.50† (14–36)		
BNIS total score	40.91 (7.397)	43 (21–49)	44.16 (3.731)	45 (32–49)	43.71 (6.198)	45† (26–50)	44.72 (3.478)	46† (35–49)	44.67 (5.626)	46† (30–50)	44.28 (3.304)	45↓ (35–49)	45.88 (3.271)	46.50† (39–50)		
DEX	26.90 (8.29)	26.00 (14–43)	27.00 (14.39)	28.00 (3–46)	24.20 (10.82)	23.50 (8–45)	29.06 (15.15)	28.50 (4–55)	25.22 (11.64)	24.50 (6–46)	28.28 (16.38)	27.00 (2–60)	29.38 (17.52)	30 (7–52)		

† Improved median results for group compared to preceding assessment.

↓ Impaired median results for group compared to preceding assessment.

± Equal median results for group compared to preceding assessment.

WAIS-III, Wechsler Adult Intelligence Scale 3rd edition, Neuropsychological Instrument; BNIS, Barrow Neurological Institute Screen for Higher Cerebral Functions; DEX, The Dysexecutive Questionnaire.

Table V. Significant changes between the assessments 1–4 (A1, A2, A3 and A4) for the Intervention Group (IG) and the Control Group (CG) and the split control group C1 and C2 (*p*-values). Significant differences between IG and CG (IG/CG).

	Test	Intervention (IG)	Control (CG)	C1	C2	IG/CG
A1–A2	Dig Span	0.003**				
	Dig Span rev	0.004**				
	Span board		0.003**			
	WM subs	0.000***	0.003**	0.011*		
	BNIS	0.006**				0.044*
A2–A3	No sig. diff.					
A1–A3	Dig Span	0.002**	0.014*			0.045*
	Dig Span rev	0.001***				0.003**
	WM subs	0.003**				
	BNIS	0.006**				
	HADS D	0.005**				
A3–A4	Dig Span rev			0.039*		
	WM subs			0.012*		
A1–A4	Dig Span			0.024*		
	Dig Span rev			0.041*		
	WM subs			0.011*		
	BNIS			0.053		
	HADS A			0.027*		
	HADS D			0.024*		

p*-value ≤ 0.05 (Sign. 2-tailed), *p*-value ≤ 0.01 (Sign 2-tailed), ****p*-value ≤ 0.001 (Sign 2-tailed).

Dig span, Digit span; rev, reversed; WM subs, Working Memory sub-scale; BNIS, Barrow Neurological Institute Screen for Higher Cerebral Functions; HADS, Hospital Anxiety and Depression Scale, D, depression, A, anxiety.

Table VI. Median values and range of the Hospital Anxiety (HADS A) and Depression (HADS D) scale at assessments 1–4 (A1, A2, A3 and A4) in the Intervention Group (IG) and the split Control Group (C1 and C2).

		HADS A	HADS D
A1	IG	6.00 (0–18)	5.00 (0–15)
A1	C1	10.50 (0–15)	9.00 (1–11)
A1	C2	6.00 (0–14)	3.00 (1–9)
A2	IG	5.50 (0–14)	4.00 (0–11)
A2	C1	10.00 (0–14)	9.50 (0–13)
A2	C2	7.50 (0–14)	3.00 (1–9)
A3	IG	5.50 (1–13)	3.00 (0–13)
A3	C1	9.00 (0–13)	7.50 (2–11)
A3	C2	5.00 (0–15)	3.00 (0–13)
A4	C1	7.00 (0–10)	6.00 (0–11)

training [17]. Cognition (BNIS) also significantly improved in the IG and tentatively in the CG after the WM training, which may suggest an improvement of general cognitive ability after computerized WM training. However, in spite of this improvement the median scores of the IG and CG for the cognitive screening test (BNIS) still remained below the cut-off during the whole study. As both groups were well-educated a cognitive performance within average before trauma could be expected on this basic cognitive screen. This suggests that the WM training may only have an impact to improve certain aspects of general cognitive function.

It is also supported by the fact that no significant improvements were found in any group (IG, C1, C2) during the period without WM training (A2–A3) when the patients

were discharged from the rehabilitation clinic (Table V). In other words, the majority of improvements found were gained only after WM training in those patients who had received the Cogmed QM. In the study design, a 3-month follow-up (A3) was included to explore if the results obtained at A2 were retained, which was shown for the IG after intervention (Table V).

Significant improvement on the WM sub-scale was found for both the IG and the CG between A1 and A2. However, for the IG that received the WM training the increase in median score was greater than the gain in median score for the CG (Table IV). Partly some change in both groups might depend on spontaneous recovery and advantages of the contemporary neurorehabilitation, but the proportionally greater gain for the IG might be contributed by the WM training. Only for the IG this improvement also remained significant at follow-up (A3) compared to baseline (A1), indicating a more robust improvement in this group over time after WM training (Table V), as is supported by other studies [10–13, 31].

The significant change relative to improvement on the BNIS found for the IG and the trend toward significance for the C1 after computerized WM training suggests that this training could have had an impact on general cognitive ability. One explanation to the specific effect of the WM program might be the continuous feedback and reinforcement of the program that could explain the stronger effect of the WM training in addition to the ordinary neurorehabilitation. The special features of the Cogmed QM training, where the demands of intensity and attention continuously adapt to the performance, might stimulate to an increased effort by the patient. The cognitive improvement (BNIS) may also possibly benefit more when starting from a lower level [13]. In this case the IG had a lower median score ($Md = 43$) at baseline on BNIS as compared to the CG ($Md = 45$). Working memory, as a key function in human cognition, is dependent on adequate dopamine neurotransmission, which can improve as a result of WM training [15]. The training of working memory, which was found to improve working memory capacity, was also reported to be associated with prefrontal and parietal changes in the density of cortical dopamine D1 receptors. McNab et al. [15] concludes that this plasticity of the dopamine D1 receptor system demonstrates a reciprocal interplay between mental activity and brain biochemistry *in vivo*. These findings might also be a result of this brain plasticity that could explain the improvements in both the IG and C1 in the BNIS after their Cogmed QM training.

Disturbed WM has been associated with a negative view of oneself and psychological health [1, 2]. During this study both the IG and CG remained, on average, below the cut-off scores defining mild anxiety and mild depression. However, after WM training fewer patients were rated as having mild, moderate or severe anxiety and/or depression in both groups (Table VI), indicating that some improvement in mood might be expected after Cogmed QM training. Patients who completed the WM training after discharge from the clinic (C1) were more depressed at A1 compared to the patients who declined the WM training (C2). After the WM training, the C1 improved significantly in both anxiety and depression, as did the IG; however, with a long-term effect in depressive mood at A3 (Table VI). This result also indicates support for

the suggestion that computerized WM training could stabilize mood and psychological health. Contrary to the results of the IG and the C1, only the C2, throughout the study, showed negative correlations between the cognitive level (BNIS) and the perception of executive deficits (DEX), suggesting better cognition associated with fewer executive problems. A possible explanation may be that, as seen at A1, more patients in the IG and C1 showed impaired awareness of their performance compared to the C2.

The strong associations between executive function, anxiety and depression indicate the importance of taking the patient's subjective experience seriously, as this is the objective reality on which he or she bases their reactions. In neurorehabilitation the combination of evident-based training methods together with neuropsychological therapy and support can confirm actual impairment and the distress it causes. Thus, it could be beneficial for the patient to work through and process the experienced trauma of the brain injury and rebuild a realistic self-awareness and quality-of-life.

Both the IG and CG reported dysexecutive problems, problems that were severe or very severe throughout the study. However, in both groups the span between high and low ratings increased over time. One explanation for this might be more diversified reactions to the impact of the injury during the rehabilitation period and the WM training. Some patients still experienced extensive remaining dysexecutive problems in daily life despite an improved cognitive ability, as seen in the IG and C1, which may be due to an increased awareness of problems. Already at baseline more patients in C2 demonstrated an awareness of their performance level, thus being more realistic. This was supported in this group also by a better cognitive function associated with fewer experienced dysexecutive problems at the follow-ups (A2 and A3). These observations may reflect that patients in the C1 had a more vulnerable personality and were more sensitive to adversity, with stronger reactions upon suffering brain damage. Returning to the clinic after discharge for computerized WM, training may have been considered an opportunity to further recovery, as well as receiving the support and security they had felt previously by re-establishing contact with the staff.

One limitation in this study was the small sample size, in that a larger sample could have given more pronounced differences between the groups. As only 38 patients completed the study, the results must be cautiously interpreted. However, it was considered that this sample was robust enough to answer the questions of this study of exploring the effect of computerised WM training on cognitive functions and mood.

The many patients who refused, aside from those that dropped out of the study, may suggest that computerised WM training might not be applicable for all due to the demands of the training programme and the motivation needed to fulfil the training. Another limitation of the study was the inability to compare the IG and CG at A4, as only eight patients in the CG completed the WM training.

For further evidence of the effect of the Cogmed QM training, follow-ups of the long-term effects at later time-points are recommended. The aim of the present study was to investigate whether patients with a dysfunctional working

memory improve working memory, cognitive function and psychological health after computerized WM training and to give some evidence to support this. However, there is a need to explore if the cognitive and psychological gains after Cogmed QM training also transfer to everyday activities. Such aspects were analysed by the authors for the same sample and the same intervention was reported in a separate article [36]. Still there is a need for further studies with larger samples and good and reliable instruments that may reflect the functional ability in meaningful activities.

In conclusion, the results of this study seems to support the use of computerized training for impaired working memory after acquired brain injury in the sub-acute phase of rehabilitation. In addition to an improvement in working memory and information processing, gains in cognition and enhanced psychological health were interpreted as benefits that the computerized training possibly added to ordinary rehabilitation. Further research, however, is needed.

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Declaration of interest

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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